

STA 2002

Lecture 6

Confidence interval

Sample size

Review

A 95% confidence interval for θ is a random interval that contains θ with probability 0.95

How to construct a confidence interval?

1. Find a point estimator (MLE or MME) and **its sampling distribution**
2. Transform the estimator such that the transformed "estimator" (approximately) follows a standard distribution and **contains the parameter of interest** θ
3. Obtain the quantile of the standard distribution---A "confidence interval" for the transformed "estimator"
4. Sort up the inequality of the transformed "estimator" to build the confidence interval of θ

Techniques

1. Not normal → Approximately normal by CLT
2. Unknown variance → sample variance as a plug-in estimator when n large
   t-distributed when n small

Confidence interval for the mean

X is normally distributed, σ is known

$$\left[\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

X is arbitrarily distributed, σ is known

$$\left[\bar{x} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \bar{x} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

X is normally distributed, σ is NOT known

$$\left[\bar{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

X is arbitrarily distributed, σ is NOT known

$$\left[\bar{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

CI for the difference between two means: Summary

Assume that X, Y are normally distributed,

- σ_x and σ_y are known $\quad [\bar{x} - \bar{y} - z_{\alpha/2}\sigma_W, \bar{x} - \bar{y} + z_{\alpha/2}\sigma_W] \quad \sigma_W = \sqrt{\sigma_X^2/n + \sigma_Y^2/m}$
- σ_x and σ_y are NOT known and n and m are large $\quad \bar{x} - \bar{y} \pm z_{\alpha/2}\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$
- σ_x and σ_y are NOT known, n and m are not large and $\sigma_x = \sigma_y$

$$\left[\bar{x} - \bar{y} - t_0 s_p\sqrt{\frac{1}{n} + \frac{1}{m}}, \bar{x} - \bar{y} + t_0 s_p\sqrt{\frac{1}{n} + \frac{1}{m}}\right] \quad \text{where } t_0 = t_{\alpha/2}(n+m-2), \; S_P = \sqrt{\frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}}$$

- σ_x and σ_y are NOT known, n and m are not large and **unequal** (Aspin, Biometrika, 1949)

$$W = \frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}} \sim t_r \text{ approximately with } r = \frac{\left(\frac{s_x^2}{n} + \frac{s_y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{s_x^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{s_y^2}{m}\right)^2}$$

CI for the difference between two means (2.3): X, Y are normally distributed, σ_x and σ_y are unknown and unequal

$$\bar{x} - \bar{y} \pm t_{\alpha/2}(r)\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$$

with a corrected degree of freedom:

$$r = \frac{\left(\frac{s_x^2}{n} + \frac{s_y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{s_x^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{s_y^2}{m}\right)^2}$$

If r is not an integer, then use the greatest integer in r

TABLE 8.1. Pig Blood Loss Data (ml)

Control Group Pigs	Treatment Group Pigs
786	543
375	666
4446	455
2886	823
478	1716
587	797
434	2828
4764	1251
3281	702
3837	1078
Sample mean = 2187.40	Sample mean = 1085.90
Sample standard deviation = 1824.27	Sample standard deviation = 717.12

Using pooled variance:

$$S_p^2 = \{S_t^2(n_t - 1) + S_c^2(n_c - 1)\}/[n_t + n_c - 2] = \{(717.12)^2\ 9 + (1824.27)^2\ 9\}/18$$

$$S_p = 1475.89$$

$$[(\bar{X}_t - \bar{X}_c) - t_0 S_p\sqrt{(1/n_t) + (1/n_c)},\ (\bar{X}_t - \bar{X}_c) + t_0 S_p\sqrt{(1/n_t) + (1/n_c)}]$$

$$= [-2082.07, -120.93]$$

Assuming different variance:

$$\frac{\{(S_t^2/n_t) + (S_c^2/n_c)\}^2}{[1/(n_c - 1)](S_c^2/n_c)^2 + [1/(n_t - 1)](S_t^2/n_t)^2} = 11.717$$

Using 11 degrees of freedom:

$$[(\overline{X}_t - \overline{X}_c) - t_0\sqrt{(S_t^2/n_t) + (S_c^2/n_c)}, (\overline{X}_t - \overline{X}_c) + t_0\sqrt{(S_t^2/n_t) + (S_c^2/n_c)}]$$

$$= [(1085.9 - 2187.4) - 2.185\sqrt{332796.1}, (1085.9 - 2187.4) + 2.185\sqrt{332796.1}]$$

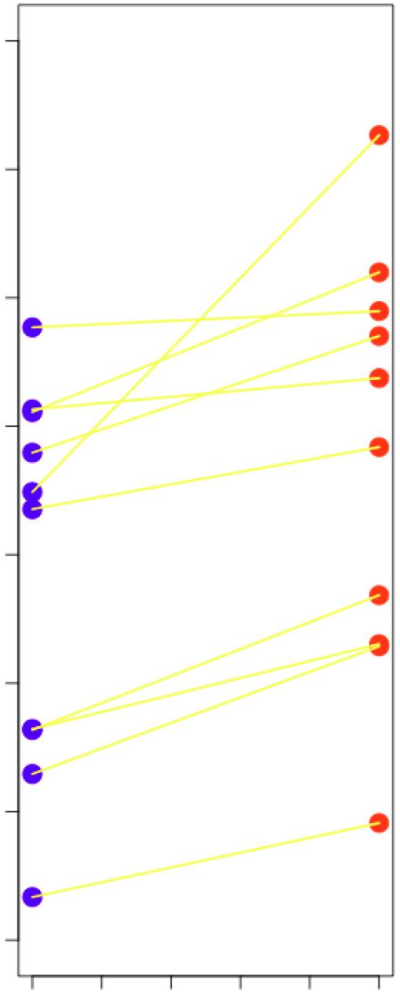
$$= [-2361.99, 158.99]$$

(A interpolated critical value $t_0 = t_{\alpha/2}(11.7)$ is used here. Let's ignore the small discrepancy)

Paired samples

Definition 8.4 Two samples are said to be **paired** when each data point in the first sample is matched and is related to a unique data point in the second sample.

Example 8.3 The paired samples may represent two sets of measurements on the same people. In this case each person is serving as his or her own control, as in Equation 8.1. The paired samples may also represent measurements on different people who are chosen on an individual basis using matching criteria, such as age and sex, to be very similar to each other.



Paired samples

In the generation of CI, the interest is the **mean of the within-pair difference**, NOT the difference in mean of the two groups.

First take difference within pair, then treat the difference as a single variable. Let $\bar{d}$ and s_d denote the observed mean and standard deviation of pairwise difference.

Confidence Interval for the True Difference (Δ) Between the Underlying Means of Two Paired Samples (Two-Sided)

A two-sided $100\% \times (1 - \alpha)$ CI for the true mean difference (Δ) between two paired samples is given by

$$\left(\bar{d} - t_{n-1,\alpha/2}\, s_d/\sqrt{n},\ \bar{d} + t_{n-1,\alpha/2}\, s_d/\sqrt{n}\right)$$

Paired samples

An experiment was conducted to compare people's reaction times to a red light versus a green light. When signaled with either the red or the green light, the subject was asked to hit a switch to turn off the light. When the switch was hit, a clock was turned off and the reaction time in seconds was recorded. The following results give the reaction times for eight subjects:

Subject	Red (x)	Green (y)	$d = x - y$
1	0.30	0.43	–0.13
2	0.23	0.32	–0.09
3	0.41	0.58	–0.17
4	0.53	0.46	0.07
5	0.24	0.27	–0.03
6	0.36	0.41	–0.05
7	0.38	0.38	0.00
8	0.51	0.61	–0.10

CI for paired samples

$\bar{d}$=-0.0625, s_d=0.0765

$$\bar{d} \pm \frac{t_{\frac{\alpha}{2}}(n-1)\, s_d}{\sqrt{n}} \rightarrow \quad [-0.1265, 0.0015]$$

CI for the difference between two means with the same variance

$\bar{x}$=0.3700, $\bar{y}$=0.4325, s_x= 0.1124, s_y= 0.1173

$$s_p = \sqrt{\frac{s_X^2 + s_y^2}{2}} = 0.1149$$

$$\bar{x} - \bar{y} \pm t_{\frac{\alpha}{2}}(2n-2)\sqrt{\frac{1}{n} + \frac{1}{n}}\, s_p \rightarrow [-0.1857, 0.0607]$$

In general, CI for paired samples will be shorter than CI for the difference between two means, because we have more information, that is, pairwise correspondence. The variance of $\bar{d}$ will be smaller than the variance of $\bar{x}$ - $\bar{y}$.

CI for proportion

$\widehat{\boldsymbol{p}}$:	$\frac{x}{n}$
s.e. of $\hat{p}$:	$\sqrt{\frac{p(1-p)}{n}}$
Distribution of $\hat{p}$:	Approximately normal, $N(p, \frac{p(1-p)}{n})$

Reminder:

X is binomial (n, p), then

E(X)=np, var(X)=np(1-p)

The $100(1-\alpha)\%$ confidence interval:

$$\frac{x}{n} \pm z_{\alpha/2}\sqrt{\frac{\frac{x}{n}\left(1-\frac{x}{n}\right)}{n}}$$

CI for proportion

Cancer Suppose we are interested in estimating the prevalence rate of breast cancer among 50- to 54-year-old women whose mothers have had breast cancer. Suppose that in a random sample of 10,000 such women, 400 are found to have had breast cancer at some point in their lives. We have shown that the best point estimate of the prevalence rate p is given by the sample proportion $\hat{p} = 400/10,000 = .040$. How can an interval estimate of the parameter p be obtained? (See the solution in Example 6.49.)

$$\hat{p} = .040 \quad \alpha = .05 \quad z_{\alpha/2} = 1.96 \quad n = 10,000$$

Therefore, an approximate 95% CI is given by

$$\left[.040 - 1.96\sqrt{.04(.96)/10,000}, .040 + 1.96\sqrt{.04(.96)/10,000}\right]$$
$$= (.040 - .004,\ .040 + .004) = (.036,\ .044)$$

CI for difference in proportion

$\widehat{p_1}$:	$\frac{x_1}{n_1}$
s.e. of $\widehat{p_1}$:	$\sqrt{\frac{p_1(1-p_1)}{n_1}}$
Distribution of $\widehat{p_1}$:	~Normal, $N(p_1, \frac{p_1(1-p_1)}{n_1})$

$\widehat{p_2}$:	$\frac{x_2}{n_2}$
s.e. of $\widehat{p_2}$:	$\sqrt{\frac{p_2(1-p_2)}{n_2}}$
Distribution of $\widehat{p_2}$:	~Normal, $N(p_2, \frac{p_2(1-p_2)}{n_2})$

$\widehat{p_1} - \widehat{p_2}$:	$\frac{x_1}{n_1} - \frac{x_2}{n_2}$
s.e. of $\widehat{p_1} - \widehat{p_2}$:	$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$
Distribution of $\widehat{p_1} - \widehat{p_2}$:	~Normal, $N(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2})$

The $100(1-\alpha)\%$ confidence interval:

$$\left(\frac{x_1}{n_1} - \frac{x_2}{n_2}\right) \pm z_{\alpha/2}\sqrt{\frac{\frac{x_1}{n_1}(1-\frac{x_1}{n_1})}{n_1} + \frac{\frac{x_2}{n_2}(1-\frac{x_2}{n_2})}{n_2}}$$

CI for difference in proportion

n_1 is 100 and n_2 is 50.

$x_1=85;\ x_2=26$

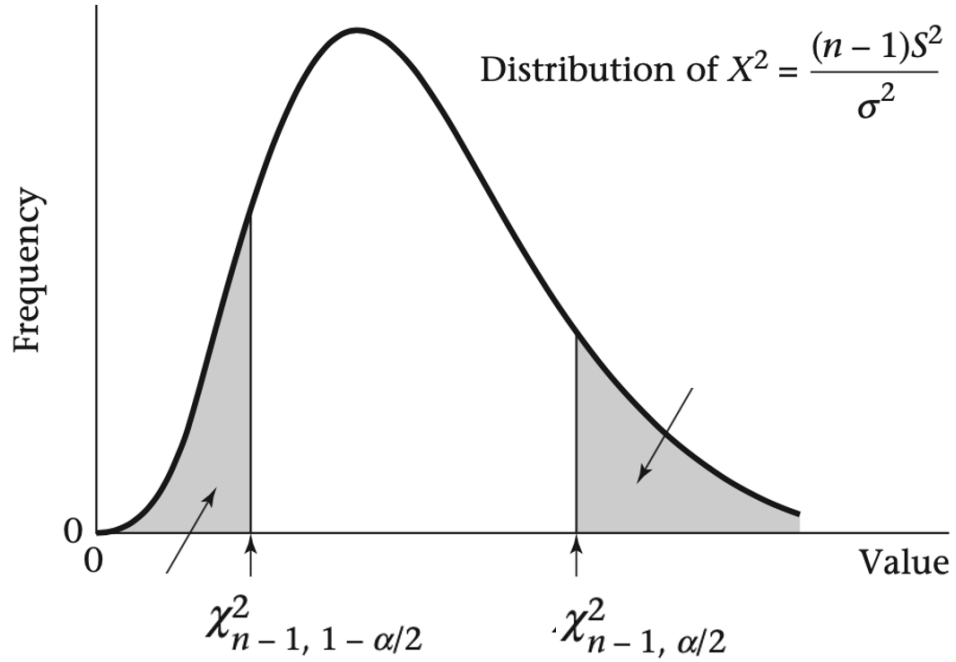
The 95% CI of the difference in proportions is:

$$[(0.85 - 0.52) - 1.96\sqrt{0.85(0.15)/100 + 0.52(0.48)/50}, (0.85 - 0.52) + 1.96\sqrt{0.85(0.15)/100 + 0.52(0.48)/50}]$$

$$= [0.175, 0.485].$$

CI for variance

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$



$$\chi^2_{n-1,\,1-\alpha/2} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{n-1,\,\alpha/2}$$

CI using the exact distribution

Not required

The existence of exact CI for proportion when normal approximation is not proper.

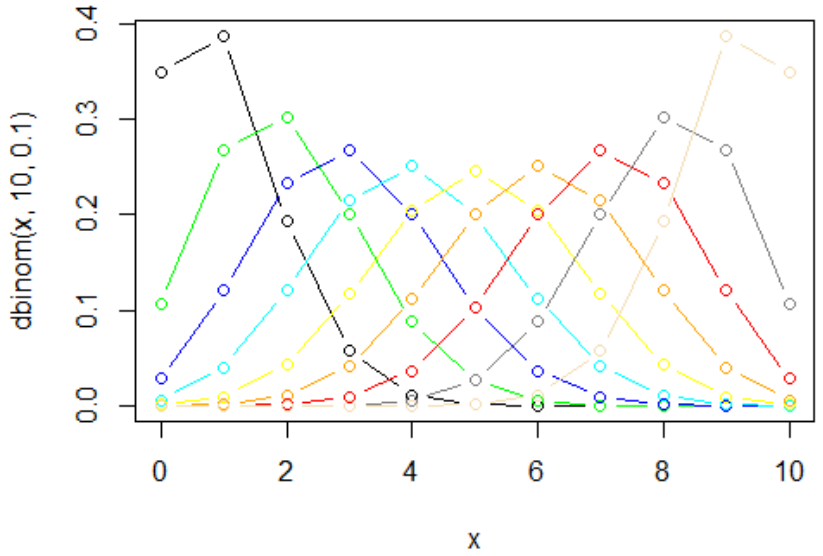
Exact Method for Obtaining a CI for the Binomial Parameter p (Clopper-Pearson Method)

An exact $100\% \times (1 - \alpha)$ CI for the binomial parameter p that is always valid is given by (p_1, p_2), where p_1, p_2 satisfy the equations

$$Pr(X \geq x \mid p = p_1) = \frac{\alpha}{2} = \sum_{k=x}^{n} \binom{n}{k} p_1^k (1 - p_1)^{n-k}$$

$$Pr(X \leq x \mid p = p_2) = \frac{\alpha}{2} = \sum_{k=0}^{x} \binom{n}{k} p_2^k (1 - p_2)^{n-k}$$

Need numerical solution. Hard to do by hand.
But is common on computer.



Sample size

In study design, increased sample size if often tied to higher cost.

If we would like to achieve a certain precision in the estimation, what is minimum the sample size needed?

At a given confidence level, the precision is reflected by the half-width of the CI.

Here (we assume) there is no bias.

Focus on precision.

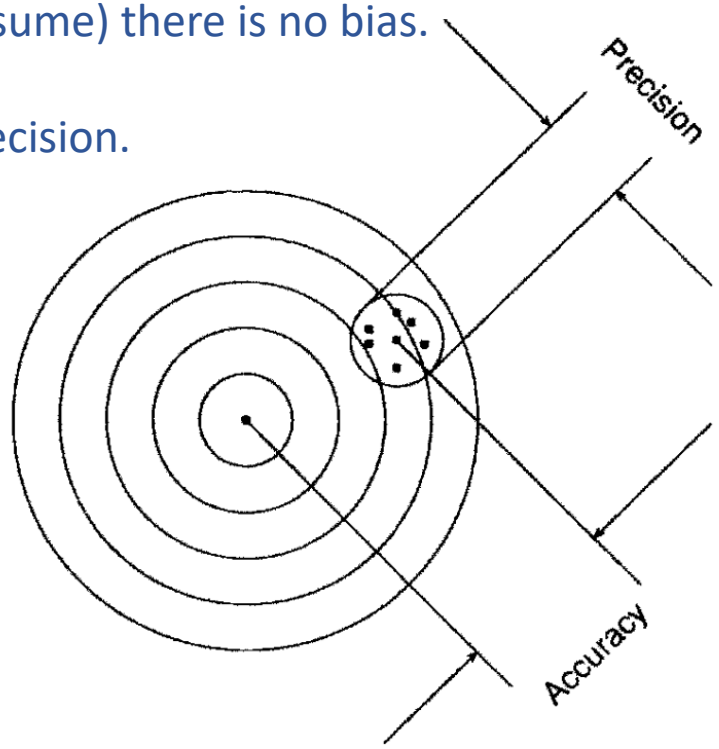


Figure 4.29 Accuracy involves the concept of bias.

Larger sample size → smaller CI

Smaller distribution variance → smaller CI

Sample size

To estimate sample size, we need to assume we know something about the distribution

- σ: Assumed, or estimated using a sample s
- As we don't know n, we can no longer find $t_{\alpha/2}(n-1)$. We use normal $z_{\alpha/2}$ to approximate

Ex:

Let us consider the case where we are sampling from a normal distribution with an assumed standard deviation of 5, and let us assume that we want the half-width of the 95% confidence interval to be no greater than 0.5.

$$1.96\sigma/\sqrt{n} \leq 0.5$$

$$3.8416(5/0.5)^2 = 384.16$$

$$n \geq 385$$

Sample size

For proportion, need to assume a "likely" p (could be estimated from previous samples).

Ex:

A particular area contains 8000 apartment units. In a preliminary survey, 12% of the respondents said they planned to sell their apartments within the next year. We want to estimate the interval for the total number of owners planning to sell. Suppose that an 95% confidence interval with a half-width of 200 is desired. Neglecting the finite population correction, how many owners do we need to survey?

$$1.96N\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \frac{5095}{\sqrt{n}} \leq 200$$

$$n \geq 649$$